

## Lab 3 - Architecture Justification

### **Architecture selected: Layered Architecture**

The Coffee Kiosk is organized into User Interface, Order Manager, Payment Service, Receipt Printer, and Database components. This separates interaction, order processing, payment, printing, and storage responsibilities.

### **Two reasons for selection:**

1. Separation of concerns: each component has a focused responsibility, making the system easier to understand and maintain.
2. Modularity: individual components can be changed or replaced with limited impact on other components.

### **Security advantage:**

The Payment Service can be isolated from the user-interface and order-processing responsibilities, reducing direct exposure of payment-related operations.

### **Performance advantage:**

Components can handle their responsibilities independently, so operations such as data access and order processing do not require unnecessary work across the whole system.